SEID 2364

Dr. David B. Smith
Assignment 2 – Defining Intelligence
Last updated 2026-03-05 – dbs
Due: Week 3

Table of Contents

Purpose 2
Part 0 Systems Setup Verification and Journal Preparation 2
0.1 GitHub Repository Setup Setup (Required) 2
0.2 Creating a GitHub personal branch (Optional) 3
0.3 Zotero 3
Part 1 Initial Speculative Definition of Intelligence 4
1.1 Copy your initial provocations into a Journal Entry 4
1.2 Redefine your definition of intelligence based on our class discussions 4
Part 2 Core Readings 5
2.1 Legg & Hutter – A Collection of Definitions of Intelligence 5
2.2 Stanford Encyclopedia of Philosophy 5
2.3 Alan Turing – Computing Machinery and Intelligence (1950) 5
2.4 Further Research and Zotero Contributions 6
Part 3 Starting to Construct a Theory of Intelligence 7
3.1 Developing a rule set 7
3.2 – Boundary Testing 7
Submission and Preparation 8
Evaluation Criteria 8
TL;DR 8

Purpose

This assignment examines the problem of defining intelligence. Throughout history, philosophers, scientists, engineers, and researchers have proposed many different definitions of intelligence. Some definitions emphasize reasoning or cognition, others emphasize learning, adaptation, prediction, environmental interaction, or goal-directed behavior.

Rather than assuming a single correct answer, this assignment treats definitions of intelligence as analytical models. A model is a map of reality: it simplifies complexity so that we can reason about systems and make decisions.

In this course we are interested in intelligence because it relates directly to agency, autonomy, and responsibility in data-driven and computational systems. As data scientists and computer scientists, you will work with systems that influence decisions, shape behavior, and interact with society. To reason ethically about such systems we must first understand what we mean when we say that a system is intelligent.

Your goal is therefore not to produce an exhaustive definition of intelligence. Instead, you will develop a working analytical model that can help distinguish intelligent from non-intelligent systems.

Your model must be capable of describing both human intelligence and machine intelligence.

The ethical frameworks you started to develop in the last assignment will remain important as we move toward questions of integration, agency, and responsibility later in the semester, even though they are not the primary focus of this assignment.

Part 0 Systems Setup Verification and Journal Preparation

Before beginning the analytical work, verify that your course infrastructure is operational.

Complete the following GitHub setup steps.

0.1 GitHub Repository Setup Setup (Required)

You **must ensure that you have access to the course GitHub repository**.
You will have already been invited to the repository—verify that you can log in and view the course materials.

At minimum, you should be able to open and clone the repository locally (or access it via GitHub Desktop or the web interface) so that you can follow along during in-class demonstrations.

Creating a personal branch and folder structure is OPTIONAL at this stage**.**

0.2 Creating a GitHub personal branch (Optional)

If you choose to complete the optional setup now:

- Create a new branch using your name
    - Example:
    - `lastname-firstname`
- Navigate to the `students` directory in the repository
- Create a new folder with the same name as your branch
    - Example: `students/lastname-firstname/`
    - Inside this folder, create a file named:
- `README.md`

Your `README.md` should include:

- your name
- your program or academic focus
- a brief statement of your interest in AI, data science, or computational systems
- a short description of what you currently think intelligence might be

Commit and push any optional changes to your branch.

Note: We will be working with GitHub together during class.
If you were not present when this assignment was previously introduced, or if you encountered issues accessing the repository, we will review this setup again.

0.3 Zotero

Ensure you also have access to the shared Zotero library for the course.

You must install both components of Zotero on your local machine:

- Zotero Desktop Application
- Zotero Browser Connector

Installers are available at: https://www.zotero.org/download/

The desktop application manages your research library, while the browser connector allows you to quickly capture sources from the web. Note that the Zotero Connector WILL NOT WORK if you do not have the local Zotero App and that it is open.

After installation, verify that you can save a reference into Zotero using the connector.

These systems will support research, collaboration, and documentation throughout the semester.

Part 1 Initial Speculative Definition of Intelligence

Before conducting deeper research, reflect on our class discussion and the provocations introduced during the lecture. This section is intended to capture your current intuitions about intelligence before they are shaped by formal readings or external sources. The goal is not to arrive at a polished or defensible theory, but to make your initial assumptions explicit so they can be examined and refined later.

1.1 Copy your initial provocations into a Journal Entry

If you have not already done this, copy EXACTLY your initial provocations from the class into your Journal.

1.2 Redefine your definition of intelligence based on our class discussions

Write a **brief speculative definition of intelligence** based on your current thinking, and from our class discussion. This section should be **short**. The goal is to capture your current intuition before you engage in deeper research. However, any definition should be able to encompass both biological and machine. Note that "intelligence" is a term of use.

This definition should represent your *initial intuition* before extensive research.

In your ethical journal, create a short section titled: **Week 2: Initial Speculative Definition of Intelligence.**

In this section, briefly state your current intuitive definition of intelligence based on your own thinking and our class discussion, prior to engaging in deeper research.

Part 2 Core Readings

Once you have established this baseline definition, it is time to engage in deeper research. The readings in this section were selected to present a wide range of different definitions of intelligence drawn from philosophy, computer science, and artificial intelligence research. Rather than converging on a single theory or framework, these texts illustrate how intelligence has been defined in diverse and sometimes incompatible ways, depending on underlying assumptions about cognition, behavior, learning, and purpose. You should read these works comparatively, focusing on how and why their definitions differ, and what those differences imply for understanding both human and machine intelligence. **You should take notes within your journal.**

2.1 Legg & Hutter – A Collection of Definitions of Intelligence

Reading link: https://arxiv.org/abs/0706.3639

Review the overview compiling more than seventy historical definitions of intelligence. Identify definitions that resonate with you.

Consider:

- What patterns appear across definitions?
- How does Legg and Hunter's proposal of "Universal Intelligence" attempt to unify these approaches.

2.2 Stanford Encyclopedia of Philosophy

Reading link: https://plato.stanford.edu/

Consult at least one relevant entry such as Intelligence, Artificial Intelligence, or Agency.

Select at least one article or section that appears most aligned with your own emerging ideas about intelligence. Use this source to help refine or challenge your developing definition.

Focus on how philosophical traditions attempt to characterize intelligence and its relationship to reasoning, learning, behavior, and cognition.

2.3 Alan Turing – Computing Machinery and Intelligence (1950)

Reading link: https://phil415.pbworks.com/f/TuringComputing.pdf

Alan Turing's 1950 paper is historically significant because it reframed intelligence as an operational and computational problem rather than a question of internal mental states. This shift strongly influenced the development of computer science, information theory, and artificial intelligence by making intelligence something that could be evaluated through computation and observable performance.

Focus specifically on the section describing the “Imitation Game” (Turing Test) as a testable evaluation framework. Then consider how Turing engages with edge conditions, handles objections, and then reaches a solid conclusion in his final section.

Consider:

- What assumptions about intelligence does this approach imply?
- What forms of computational intelligence might not pass the imitation game, and why?
- Does this test implicitly treat human-like behavior as the benchmark for intelligence? Why or why not?
- Might there be a different configuration or other types of tests that would be equally valid?

2.4 Further Research and Zotero Contributions

Intelligence is one of the most widely discussed concepts in science, philosophy, and engineering. The readings above represent only a small portion of the literature.

You are required to locate additional scholarly or technical sources discussing intelligence, machine intelligence, or artificial intelligence.

Do not rely solely on the sources provided in this assignment.

Using academic databases, libraries, or reputable research sources, identify **additional articles, papers, or essays** related to intelligence.

You must contribute **at least one source** to the shared course Zotero library.

Zotero Library Link: https://www.zotero.org/groups/6444008/seid_2364_ethics/library

For each source you contribute:

- Ensure the bibliographic information is complete
- Attach the article or link if available
- Add appropriate tags or notes describing the topic

This process will help us collaboratively build a shared research base for the course.

Part 3 Starting to Construct a Theory of Intelligence

Using the readings from this assignment, your original speculations, and any additional sources you explored, develop your own analytical definition of intelligence. Rather than attempting to synthesize all perspectives, identify one or more theories, models, or approaches to intelligence that resonate with you and that meaningfully influence how you understand what intelligence is and how it operates.

3.1 Developing a rule set

Your definition must be **expressed as a set of explicit rules or conditions** that determine whether intelligence is present. These rules should be clear enough that another person could apply them to evaluate a system. Your model must be applicable to both biological and machine systems, while also recognizing that certain aspects of intelligence may manifest differently in these two contexts.

Think of this task as **constructing a classification framework** rather than writing a philosophical essay. Assume that intelligence exists on a continuum rather than as a simple binary property. If intelligence is scalar, an important analytical question follows: **where, in your model, does intelligence transition from non-intelligent to intelligent**? Your rules should attempt to address this boundary.

Be prepared to explain how intelligence differs from agency in your model, and why certain systems may exhibit one without the other.

3.2 – Boundary Testing

Once your definition is constructed, test it against several boundary cases.

Select at least:

- Two systems you believe are intelligent
- Two systems you believe are not intelligent

Explain how your rules correctly classify each system.

If your rules produce ambiguous or contradictory results, discuss why. Note, we do not expect that at this time your definition is completely consistent or exhaustive: this will be an ongoing process that we will continue to refine throughout the semester.

Submission and Preparation

Add this work as a new entry in your Ethical Journal titled: ***Journal Entry Week 2 – Defining Intelligence***

Bring your completed journal entry to the next class session.

During the following class we will:

- compare definitions across the class
- examine boundary conditions for intelligence
- discuss how intelligence relates to agency, autonomy, and responsibility

Evaluation Criteria

You are evaluated on:

- engagement with the readings and additional research
- contribution of sources to the Zotero research library
- logical clarity of your proposed rules
- ability to distinguish intelligence from non-intelligence
- recognition of boundary cases and limitations
- precision of reasoning

Agreement with any definition or set of definitions of intelligence is NOT evaluated. We also do not necessarily need to propose a UNIVERSAL definition of intelligence, but an operational one.

Brief ethical reflections are evaluated for thoughtfulness and clarity, not for the adoption of any particular ethical framework or position.

The goal of this assignment is analytical clarity and intellectual exploration rather than consensus.

TL;DR

- You already grounded ethics solidly in Assignment 1.
- Assignment 2 should not re-teach ethics, only connect definitions of intelligence to ethical consequences.
- You're doing that appropriately with a brief ethical reflection at the end of Part IV.
- Ethics is memorialized (acknowledged as foundational) but not foregrounded yet.

- The Evaluation Criteria only need one clarifying sentence to prevent students from over-weighting ethics.
- Result: continuity, reduced redundancy, no added workload, and a clear runway toward later discussions of integration, agency, and responsibility.